

SPRING BOOT ANNOTATION GUIDE

01. STEREOYPE & COMPONENT

<code>@Controller</code>	Marks class as a web controller.
<code>@RestController</code>	<code>@Controller</code> + <code>@ResponseBody</code> (returns JSON/data).
<code>@Service</code>	Logic layer; bridge between Controller & Repo.
<code>@Component</code>	Generic Spring-managed component.

02. WEB & REST API

<code>@GetMapping</code>	Handles HTTP GET (Read operations).
<code>@PostMapping</code>	Handles HTTP POST (Create operations).
<code>@PathVariable</code>	Extracts data from URL path (/id).
<code>@RequestBody</code>	Maps HTTP body (JSON) to Java object.
<code>@CrossOrigin</code>	Enables CORS for external domain access.

03. DATA ACCESS & JPA

<code>@Entity</code>	Maps Java class to a DB table.
<code>@Id</code>	Primary key for the entity.
<code>@Transactional</code>	Ensures operation atomicity (Commit or Rollback).
<code>@Query</code>	Define custom SQL/JPQL in repository.

04. VALIDATION RULES

<code>@Valid</code>	Triggers validation of an annotated object.
<code>@NotBlank</code>	Ensures string is not null or whitespace.
<code>@Email</code>	Verifies valid email address format.
<code>@Size</code>	Validates length (min/max) of value.

05. CONFIGURATION & DI

<code>@Configuration</code>	Indicates a source of bean definitions.
<code>@Bean</code>	Method-level annotation to create a bean.
<code>@Autowired</code>	Automatic dependency injection.

06. EXCEPTION HANDLING

<code>@ControllerAdvice</code>	Centralized exception handling for all controllers.
<code>@ExceptionHandler</code>	Handles specific exceptions for methods.

REMINDER: Use `@PathVariable` for resources (e.g. /users/5) and `@RequestParam` for filters (e.g. /users?role=admin).